

Module 1: Questions and Answers

1. How does AI differ from traditional programming?

Traditional programming uses rules and instructions written manually by a programmer. AI systems can learn patterns from data and use them to make decisions or predictions. In simple words, traditional programming follows predefined rules, while AI can learn from data.

2. What are some real-world applications of AI?

AI is used in many areas such as healthcare, finance, recommendation systems, autonomous vehicles, virtual assistants, chatbots, image recognition, cybersecurity, and e-commerce.

3. Explain the difference between Narrow AI, General AI and Super AI.

- Narrow AI: Designed to perform a specific task, such as voice assistants or recommendation systems.
- General AI: A theoretical AI that can perform different intellectual tasks like a human.
- Super AI: A theoretical form of AI that would surpass human intelligence in almost every area.

4. Explain how Super AI could potentially surpass human intelligence.

Super AI could surpass human intelligence by learning independently and performing reasoning, creativity, decision-making, and problem-solving faster and more effectively than humans. It could also apply knowledge across different fields without human assistance.

5. What ethical concerns are associated with Super AI?

Super AI could create concerns related to human control, safety, regulation, and alignment with human values. If such systems become highly autonomous, controlling their decisions and goals could become difficult.

6. What is Reactive Machine AI? Give examples.

Reactive Machine AI is the simplest type of AI. It responds only to the current input and does not store past experiences or learn from previous events.

Examples: IBM's Deep Blue and Google's AlphaGo.

7. Explain the concept of Theory of Mind AI.

Theory of Mind AI is an advanced concept where AI would understand human emotions, beliefs, intentions, and desires. It could change its responses based on a person's emotional or mental state.

8. How does Limited Memory AI differ from Reactive AI?

Reactive AI only responds to the current situation and has no memory of past experiences. Limited Memory AI can use historical or recent data for a short period to improve its decisions and predictions.

9. How does Machine Learning relate to Artificial Intelligence?

Machine Learning is a part of Artificial Intelligence. It allows computers to learn patterns from data and improve their performance without being explicitly programmed with every rule.

10. Give examples of Machine Learning applications in daily life.

Some common examples are spam email filtering, Netflix and YouTube recommendations, virtual assistants, chatbots, online shopping recommendations, fraud detection, image recognition, and social media feeds.

11. What is the difference between training and testing data?

Training data is used to teach a machine learning model and help it learn patterns. Testing data is new data used to check how well the trained model performs on unseen examples.

12. Which learning method is best suited for labeled data?

Supervised Learning is best suited for labeled data because both the input and the correct output are provided during training.

13. Compare Supervised and Unsupervised Learning.

Supervised Learning uses labeled data and learns the relationship between inputs and known outputs. Unsupervised Learning uses unlabeled data and finds hidden patterns or groups without predefined outputs.

14. Explain the difference between classification and regression.

Classification is used to place data into predefined categories, such as spam or not spam. Regression is used to predict a continuous numerical value, such as the price of a car.

15. What is overfitting in supervised learning?

Overfitting happens when a machine learning model learns the training data too closely instead of learning general patterns. As a result, the model may perform well on training data but poorly on new, unseen data.

16. How does unsupervised learning differ from supervised learning?

Supervised learning learns from labeled data where the correct output is provided. Unsupervised learning works with unlabeled data and tries to discover patterns, groups, or structures by itself.

17. What is dimensionality reduction?

Dimensionality reduction is a technique used to simplify a dataset by reducing the number of features while keeping important information. It can also help represent complex data in 2D or 3D for easier visualization. Examples include PCA, Kernel PCA, LLE, and t-SNE.

18. How does reinforcement learning differ from supervised learning?

Supervised learning learns from labeled examples with known correct outputs. Reinforcement learning learns through interaction with an environment by taking actions and receiving rewards or penalties.

19. Explain the concept of “reward” in RL.

A reward is feedback given to an AI agent after it takes an action. It indicates whether the action was good or bad for achieving the goal. The agent tries to learn actions that maximize the total rewards over time.

20. What are some challenges in unsupervised learning?

Some challenges include finding meaningful patterns without labeled outputs, deciding the correct number of groups, interpreting the discovered patterns, and handling complex or high-dimensional data. Dimensionality reduction and clustering are commonly used to help discover useful structures in such data.